



REVIEW

How do mRNA vaccines work?

REFERENCES	TOKENS	VERIFICATION	COMPILED
47 verified	≈9.6k	Crossref	26 Aug 2026

ABSTRACT

Messenger RNA vaccines use a lipid nanoparticle to deliver a short-lived genetic recipe for a pathogen protein, not the pathogen itself. Cells translate that recipe in the cytoplasm, and the resulting antigen trains neutralizing antibodies, memory B cells, helper T cells and killer T cells. The same formulation must balance two competing requirements: it must protect and express the RNA while provoking enough innate signalling to build immunity without suppressing translation. This balance enabled high initial efficacy against ancestral SARS-CoV-2 and protection against respiratory syncytial virus, but it does not make performance uniform across pathogens or immune contexts. Protection against infection wanes faster than protection against severe disease, variants can escape a previously matched antigen, and intramuscular vaccination produces weaker airway than blood immunity. Safety surveillance shows that serious events are uncommon, while myocarditis is a real, concentrated risk that varies by age, sex, dose and product. Manufacturing remains rapid and programmable but formulation-specific, and newer self-amplifying and circular RNA designs differ sharply in clinical maturity. The platform is therefore best understood as a tunable delivery-and-immunity system rather than a single fixed vaccine technology.

01 Introduction

Messenger RNA (mRNA) vaccination separates the information that defines an antigen from the machinery that manufactures and delivers it. The sequence can be redesigned while much of the production platform remains stable, but the biological result depends on more than sequence. Delivery into the cytoplasm, innate sensing, lymph-node organization, the anatomical site of immunity and the match between antigen and circulating pathogen all shape protection.

This creates the central tension of the platform: programmability makes the message easy to change, yet formulation and immune context determine what that message can accomplish. The sections below follow that tension from lipid-particle entry and RNA chemistry through immune memory, clinical effectiveness, waning, safety, manufacturing and next-generation RNA formats.

02 Cytoplasmic delivery makes the antigen programmable

The payload is mRNA enclosed in a lipid nanoparticle (LNP). The particle protects the RNA and releases it into the cytoplasm, where ribosomes translate it; the RNA neither enters the nucleus nor becomes DNA. Pardi & Krammer 2024, Chaudhary et al. 2021, Hou et al. 2021 Ionizable lipids bind RNA during particle formation but are closer to neutral in the body. Release from the endosome nevertheless remains inefficient, and its mechanism is still disputed. Cullis & Felgner 2024, Chatterjee et al. 2024

What the cell makes is defined by the encoded sequence rather than by a generic vaccine product. COVID-19 vac-

cines encoded prefusion-stabilized spike, building on earlier coronavirus structural work, so a validated production platform could accept a redesigned sequence. Corbett et al. 2020, Pardi & Krammer 2024 Programmability begins with the message, but it only becomes useful when the carrier delivers enough intact RNA.

03 RNA chemistry balances translation and innate alarm

Unmodified RNA activates innate sensors. Substituting modified nucleosides reduced signalling through Toll-like receptors 3, 7 and 8 and reduced cytokine release from human dendritic cells. Removing double-stranded RNA contaminants then eliminated detectable interferon and cytokine induction in tested cells and increased translation **10- to 1,000-fold**. Karikó et al. 2005, Karikó et al. 2011 The message is therefore engineered not only for what it encodes, but also for how conspicuously the innate immune system sees it.

The carrier supplies part of the alarm that modified RNA suppresses. In mice, one LNP formulation acted as an adjuvant through its ionizable lipid and interleukin-6. Another mouse study linked the RNA-sensing protein MDA5 and its type-I-interferon signal to killer-T-cell expansion while finding several proposed pathways dispensable. These mechanisms may not generalize across LNP formulations or from mice to humans. Alameh et al. 2021, Li et al. 2022, Verbeke et al. 2022, Locci & Pardi 2026

Chemical tuning also creates trade-offs. N1-methylpseudouridine reduced innate recognition but enabled +1 ribosomal frameshifting and immune recognition of off-target products in mice and people. Recoding synonymous

codons lowered the signal, and no adverse clinical outcome was established. [Mulroney et al. 2024](#) The same design choices that improve translation can therefore introduce new molecular effects that require direct measurement.

04 Lymph nodes turn a brief signal into memory

Early human trials found neutralizing antibodies together with antiviral type-1 helper and killer T-cell responses. Killer T cells expanded after the first dose, before neutralizing antibodies had fully matured. [Sahin et al. 2020](#), [Oberhardt et al. 2021](#) Serial lymph-node sampling then detected spike-specific germinal-centre B cells for at least **12 weeks** and T-follicular-helper cells for about **6 months**. Across 1,540 clones, somatic mutation rose **3.5-fold**, accompanied by affinity-matured memory B cells and marrow plasma

cells, although these cohorts were small. [Turner et al. 2021](#), [Mudd et al. 2022](#), [Kim et al. 2022](#)

The anatomical distribution of that memory is uneven. Systemic vaccination produced strong blood memory but weaker airway immunity: vaccinated participants had lower bronchoalveolar neutralization and no detectable spike-specific B or T cells there, unlike convalescents. This small study helps explain breakthrough infection but does not quantify how much the airway gap contributes. [Tang et al. 2022](#) The delivery-to-memory pathway and the boundary between human and mouse evidence are summarized in [Figure 1](#). Durable systemic memory is real, but it does not guarantee equally durable protection at every anatomical site.

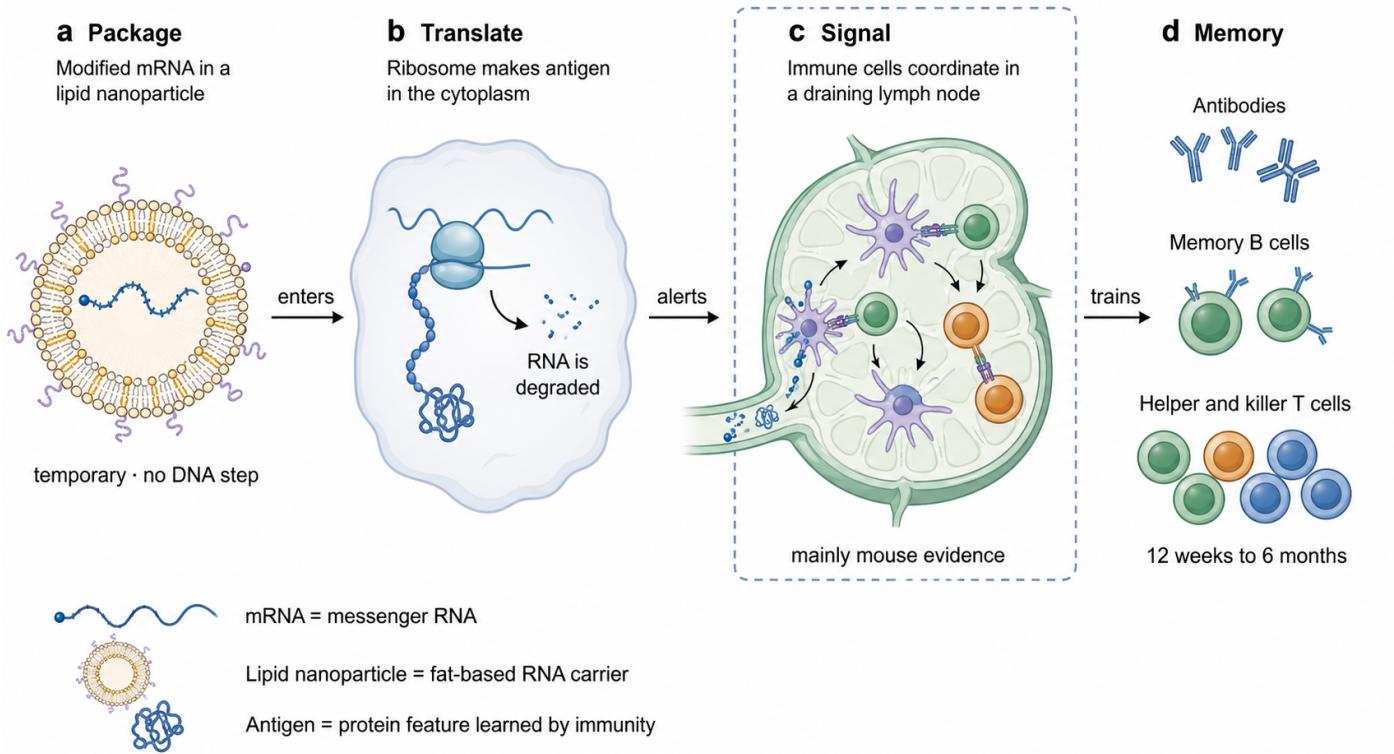


Figure 1. A temporary RNA recipe builds immune memory. An LNP delivers modified mRNA, ribosomes make antigen in the cytoplasm, immune cells co-ordinate in a draining lymph node, and antibodies plus memory B and T cells remain. Delivery and persistent human lymph-node responses are observed; the detailed sensing pathway and LNP adjuvant effect remain formulation-specific and rely heavily on mouse experiments. [Pardi & Krammer 2024](#), [Locci & Pardi 2026](#), [Alameh et al. 2021](#), [Li et al. 2022](#), [Kim et al. 2022](#)

05 Clinical performance depends on the target

VACCINE	EVIDENCE	MAIN RESULT	IMPORTANT BOUNDARY
BNT162b2	Placebo-controlled randomized trial, n=43,448	95.0% efficacy against symptomatic COVID-19	Ancestral-virus era Polack et al. 2020
mRNA-1273	Placebo-controlled randomized trial, n=30,420	94.1% efficacy; 30 severe cases, all placebo	Ancestral-virus era Baden et al. 2021
mRNA-1345 RSV	Placebo-controlled randomized trial, n=35,541, age ≥60	83.7% against lower-respiratory disease with ≥2 signs	Median follow-up 112 days Wilson et al. 2023
mRNA-1010 influenza	Two active-controlled phase-3 trials, >14,000	Stronger A but weaker B profile than licensed vaccine	Original formulation; further optimization needed Kandinov et al. 2025

The clearest early COVID-19 results showed what a well-matched antigen and formulation could achieve. Before major immune-escape variants, pivotal placebo-controlled trials reported **95.0%** efficacy (95% credible interval 90.3–

97.6; n=43,448) for BNT162b2 and **94.1% efficacy** (95% confidence interval (CI) 89.3–96.8; n=30,420) for mRNA-1273 against symptomatic COVID-19. Polack et al. 2020, Baden et al. 2021 Neutralizing antibodies tracked protection: estimated mRNA-1273 efficacy rose from 78% (95% CI 54–89) at a neutralization titre of 10 to 96% (94–98) at 1,000, while a 24-study model linked variant neutralization loss to lower protection. Correlation does not establish that antibodies mediate every component of protection. Gilbert et al. 2022, Cromer et al. 2022

Success extended to respiratory syncytial virus (RSV), where the platform protected older adults from disease and day-29 neutralization predicted lower risk. Influenza trials were less uniform, producing stronger influenza-A but weaker influenza-B responses than a licensed vaccine. Wilson et al. 2023, Ma et al. 2025, Kandinov et al. 2025 The platform is reusable; its effectiveness is not interchangeable across antigens.

06 Protection wanes fastest against infection

The same context dependence appears over time. Across 78 estimates, effectiveness fell **21.0 percentage points** (95% CI 13.9–29.8) against infection from month 1 to month 6, compared with **10.0 points** (6.1–15.4) against severe disease. Only 3 of 18 included studies had low overall risk of bias, and all predated widespread Omicron. Feikin et al. 2022 Updated vaccines restored protection without ending

the pattern. A 28-study synthesis found that bivalent boosters added **28–31% relative effectiveness** against infection and **58–62%** against severe outcomes. For XBB.1.5 vaccines, estimates fell from month 1 to month 5 from 52.9% to 26.7% against infection, 64.4% to 52.3% against hospitalization and 77.3% to 69.4% against death. Song et al. 2024, Ma et al. 2026

Estimates also depend on study population and design. A target-trial emulation of 587,137 matched pairs, mostly older men, estimated lower 120-day effectiveness of 3.15% against infection, 25.25% against hospitalization and 30.25% against death. Residual confounding and population differences probably explain part of the gap between estimates. Ioannou et al. 2025 Variant mismatch and immune history add a biological source of variation: BQ.1.1 and XBB.1 substantially escaped neutralization after a BA.5-bivalent dose, while XBB exposure often recalled B cells recognizing epitopes shared with the ancestral strain and recruited few truly XBB-specific clones in some people. Kurhade et al. 2023, Johnston et al. 2024

The outcome-specific declines and their supported, but not quantitatively partitioned, explanations are separated in Figure 2. The same systemic-versus-airway distinction seen in immune memory helps explain why protection from infection is less durable than protection from severe disease, but no single mechanism accounts for the full decline.

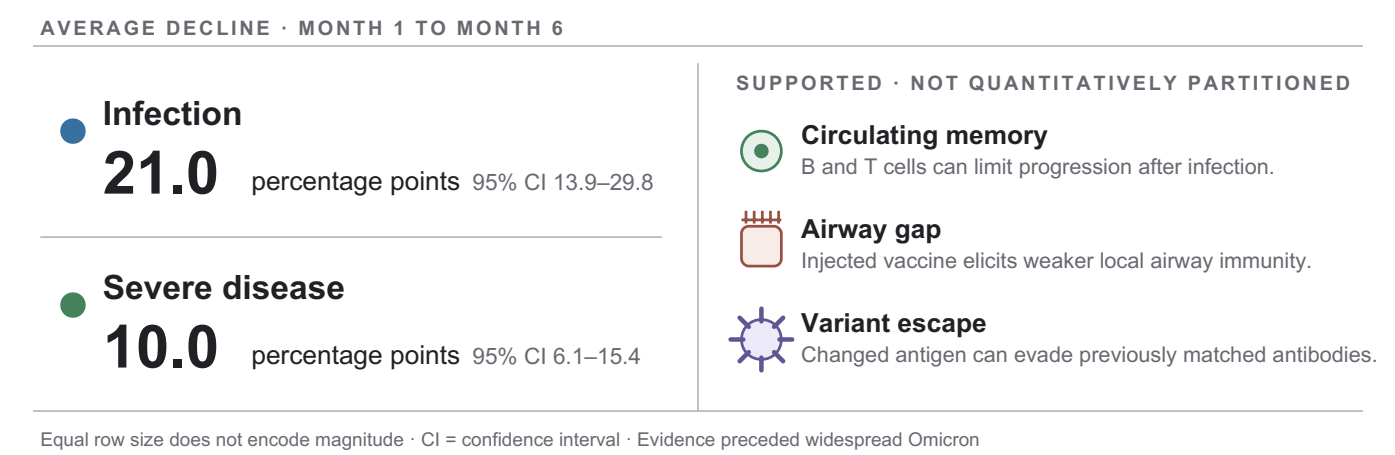


Figure 2. Protection wanes faster against infection than severe disease. Average decline from month 1 to month 6 was 21.0 percentage points against infection and 10.0 points against severe disease in a pre-Omicron synthesis. Equal row sizes avoid a false magnitude encoding; airway immunity, immune memory and variant escape are supported explanations, but the literature does not assign a percentage to each. Feikin et al. 2022, Tang et al. 2022, Johnston et al. 2024, Ma et al. 2026

07 Myocarditis risk is rare and concentrated

Large surveillance studies found that serious adverse outcomes were uncommon overall while identifying a specific myocarditis pattern. Active surveillance of **11.8 million doses** found no prespecified serious-outcome signal and anaphylaxis at about **5 per million doses**. A separate matched national study found that most examined events were not elevated but estimated 2.7 excess myocarditis cases per 100,000 vaccinated people. Klein et al. 2021,

Barda et al. 2021 A 99-million-person study reproduced myocarditis and pericarditis signals, although comparisons depend on background rates. In adolescents, pooled myopericarditis was **43.5 per million doses** (95% CI 30.8–61.6): 66.0 in males versus 10.1 in females, and 60.4 after dose 2 versus 16.6 after dose 1. Faksova et al. 2024, Guo et al. 2023

Product and dose further concentrate risk. Nordic registers estimated **55.5 excess cases per million** after BNT162b2

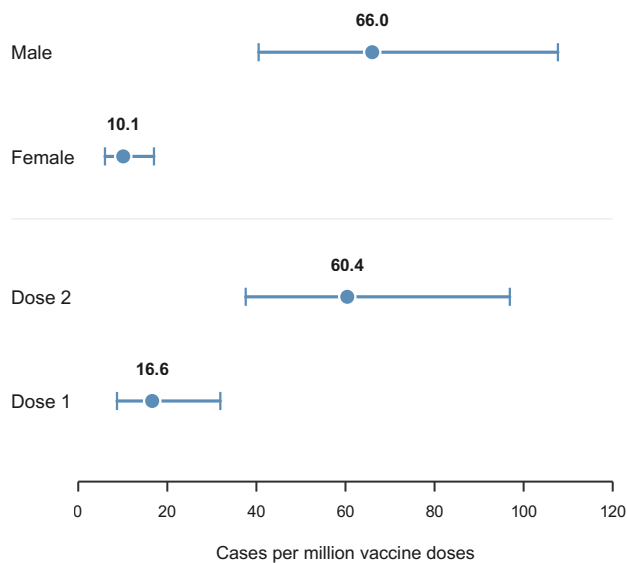
dose 2 and **183.9 per million** after mRNA-1273 dose 2 in males aged 16–24. Booster risks were lower, at 8.6 and 19.5 per million young males respectively, but remained elevated. [Karlstad et al. 2022](#), [Hviid et al. 2024](#) Infection is not a uniform comparator: across all ages it carried higher myocarditis risk, but among men younger than 40 the estimated excess after mRNA-1273 dose 2 was 97 per million,

compared with 16 per million after a positive test. [Patone et al. 2022](#)

The valid within-panel comparisons and incompatible cross-panel denominators are shown in Figure 3. The safety evidence repeats the review's wider pattern: a platform-wide average can obscure important formulation and population differences.

a Pooled adolescents, age 12–17

15 observational studies · 39.6 million doses

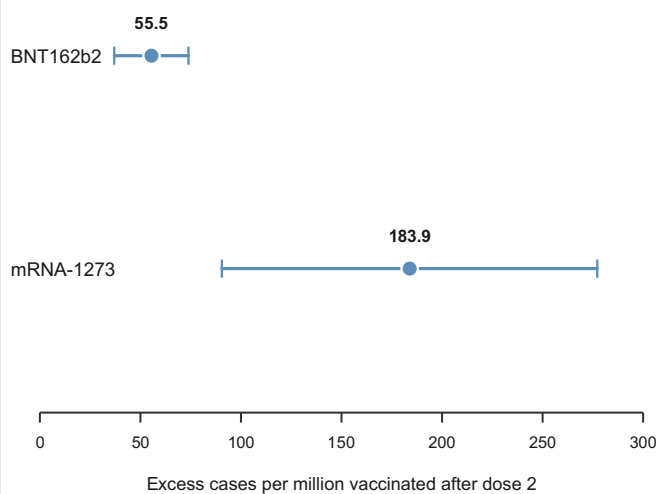


Panel a = observed cases per million vaccine doses
Sex and dose are separate pooled subgroup analyses

b Nordic males, age 16–24

Four linked national cohorts · 28-day window

Do not compare magnitudes across panels



Panel b = excess cases per million vaccinated people
95% CI = confidence interval

Figure 3. Myocarditis risk is concentrated rather than uniform. Panel a separates pooled adolescent rates by sex and dose; panel b separates Nordic excess events in males aged 16–24 by vaccine product. Populations, denominators and axes differ, so comparisons are valid within a panel but marker positions cannot be compared across panels. [Guo et al. 2023](#), [Karlstad et al. 2022](#), [Patone et al. 2022](#), [Faksova et al. 2024](#)

08 Manufacturing is fast but formulation-specific

Large-scale production couples in-vitro transcription to DNA removal, purification, particle mixing, sterile filtration and product characterization. Double-stranded RNA control, mixing reproducibility and scale-up remain bottlenecks. [Rosa et al. 2021](#), [Hou et al. 2021](#) Freeze-drying preserved one mRNA-LNP formulation for 12 weeks at room temperature and 24 weeks at 4 °C in preclinical tests. Continuous drying maintained transfection for at least 12 weeks at 4, 22 and 37 °C, but buffer, cryoprotectant and the ionizable-lipid:mRNA ratio determined whether a formulation survived drying. [Muramatsu et al. 2022](#), [Meulewaeter et al. 2023](#), [Lamoot et al. 2023](#)

Lipid chemistry also matters during liquid storage. Piperidine-based ionizable lipids reduced reactive aldehydes, RNA-lipid adducts and functional loss at refrigeration temperat-

ure. These studies identify possible routes to thermostability, not a universal room-temperature shelf life. [Hashiba et al. 2024](#) Rapid sequence exchange does not remove the need to validate the complete formulation and process.

09 New RNA formats differ in clinical maturity

[Self-amplifying RNA](#) carries replication machinery that copies the antigen template inside cells. A 5-µg ARCT-154 regimen showed 56.6% efficacy (95% CI 48.7–63.3) against any Delta-era COVID-19 and 95.3% (80.5–98.9) against severe disease. These results do not establish superiority across variants or over time. [Hø et al. 2024](#), [Pardi & Krammer 2024](#) As a fourth-dose booster (n=828), ARCT-154 produced non-inferior ancestral-strain neutralization and higher BA.4/5 neutralization than BNT162b2 at day 28. The trial measured immune responses, not disease prevention or durability. [Oda et al. 2024](#)

Circular RNA may extend expression without a replicase. An optimized Zika circular-RNA vaccine produced durable single-dose protection without obvious dengue enhancement in mice, but human safety and efficacy remain untested. Liu et al. 2024 Self-amplification has therefore crossed into clinical efficacy testing, whereas circular RNA remains a preclinical prospect.

10 Conclusion

The cross-cutting pattern is that mRNA is programmable, but the biological product is the whole sequence-formulation-host system. Cytoplasmic delivery and chemical tuning make strong antigen expression possible; lymph nodes convert that brief signal into durable systemic memory; and

clinical trials show that the platform can protect against more than one pathogen. The same evidence also defines its limits. Airway immunity is weaker than blood memory; protection from infection declines before protection from severe disease; antigenic drift can outpace the match; myocarditis risk is concentrated in particular groups and products; and stability gains do not automatically transfer between formulations. Self-amplifying RNA has clinical evidence, while circular RNA still requires human testing. Future progress therefore depends less on proving that mRNA can work than on optimizing each linked component and measuring performance in the population, pathogen and time horizon for which a vaccine is intended.

References

- Alameh MG, Tombácz I, Bettini E, Lederer K, Ndeupen S, Sittplangkoon C, Wilmore JR, Gaudette BT, Soliman OY, Pine M, Hicks P, Manzoni TB, Knox JJ, Johnson JL, Laczkó D, Muramatsu H, Davis B, Meng W, Rosenfeld AM, Strohmeier S, Lin PJC, Mui BL, Tam YK, Karikó K, Jacquet A, Krammer F, Bates P, Cancro MP, Weissman D, Luning Prak ET, Allman D, Igyártó BZ, Locci M, Pardi N (2021) Lipid nanoparticles enhance the efficacy of mRNA and protein subunit vaccines by inducing robust T follicular helper cell and humoral responses. *Immunity*. <https://doi.org/10.1016/j.immuni.2021.11.001>
- Baden LR, El Sahly HM, Essink B, Kotloff K, Frey S, Novak R, Diemert D, Spector SA, Roupheal N, Creech CB, McGettigan J, Khetan S, Segall N, Solis J, Brosz A, Fierro C, Schwartz H, Neuzil K, Corey L, Gilbert P, Janes H, Follmann D, Marovich M, Mascola J, Polakowski L, Ledgerwood J, Graham BS, Bennett H, Pajon R, Knightly C, Leav B, Deng W, Zhou H, Han S, Ivarsson M, Miller J, Zaks T (2021) Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2035389>
- Barda N, Dagan N, Ben-Shlomo Y, Kepten E, Waxman J, Ohana R, Hernán MA, Lipsitch M, Kohane I, Netzer D, Reis BY, Balicer RD (2021) Safety of the BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Setting. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2110475>
- Chatterjee S, Kon E, Sharma P, Peer D (2024) Endosomal escape: A bottleneck for LNP-mediated therapeutics. *Proceedings of the National Academy of Sciences*. <https://doi.org/10.1073/pnas.2307800120>
- Chaudhary N, Weissman D, Whitehead KA (2021) mRNA vaccines for infectious diseases: principles, delivery and clinical translation. *Nature Reviews Drug Discovery*. <https://doi.org/10.1038/s41573-021-00283-5>
- Corbett KS, Edwards DK, Leist SR, Abiona OM, Boyoglu-Barnum S, Gillespie RA, Himansu S, Schäfer A, Ziawo CT, DiPiazza AT, Dinnon KH, Elbashir SM, Shaw CA, Woods A, Fritch EJ, Martinez DR, Bock KW, Minai M, Nagata BM, Hutchinson GB, Wu K, Henry C, Bahl K, Garcia-Dominguez D, Ma L, Renzi I, Kong WP, Schmidt SD, Wang L, Zhang Y, Phung E, Chang LA, Loomis RJ, Altaras NE, Narayanan E, Metkar M, Presnyak V, Liu C, Louder MK, Shi W, Leung K, Yang ES, West A, Gully KL, Stevens LJ, Wang N, Wrapp D, Doria-Rose NA, Stewart-Jones G, Bennett H, Alvarado GS, Nason MC, Ruckwardt TJ, McLellan JS, Denison MR, Chapell JD, Moore IN, Morabito KM, Mascola JR, Baric RS, Carfi A, Graham BS (2020) SARS-CoV-2 mRNA vaccine design enabled by prototype pathogen preparedness. *Nature*. <https://doi.org/10.1038/s41586-020-2622-0>
- Cromer D, Steain M, Reynaldi A, Schlub TE, Wheatley AK, Juno JA, Kent SJ, Triccas JA, Khoury DS, Davenport MP (2022) Neutralising antibody titres as predictors of protection against SARS-CoV-2 variants and the impact of boosting: a meta-analysis. *The Lancet Microbe*. <https://doi.org/10.1016/s2666-5247%2821%2900267-6>
- Cullis PR, Felgner PL (2024) The 60-year evolution of lipid nanoparticles for nucleic acid delivery. *Nature Reviews Drug Discovery*. <https://doi.org/10.1038/s41573-024-00977-6>
- Faksova K, Walsh D, Jiang Y, Griffin J, Phillips A, Gentile A, Kwong JC, McCartney K, Naus M, Grange Z, Escolano S, Sepulveda G, Shetty A, Pillsbury A, Sullivan C, Naveed Z, Janjua NZ, Giglio N, Perälä J, Nasreen S, Gidding H, Hovi P, Vo T, Cui F, Deng L, Cullen L, Artama M, Lu H, Clothier HJ, Batty K, Paynter J, Petousis-Harris H, Buttery J, Black S, Hviid A (2024) COVID-19 vaccines and adverse events of special interest: A multinational Global Vaccine Data Network (GVDN) cohort study of 99 million vaccinated individuals. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2024.01.100>
- Feikin DR, Higdon MM, Abu-Raddad LJ, Andrews N, Araos R, Goldberg Y, Groome MJ, Huppert A, O'Brien KL, Smith PG, Wilder-Smith A, Zeger S, Deloria Knoll M, Patel MK (2022) Duration of effectiveness of vaccines against SARS-CoV-2 infection and COVID-19 disease: results of a systematic review and meta-regression. *The Lancet*. <https://doi.org/10.1016/s0140-6736%2822%2900152-0>
- Gilbert PB, Montefiori DC, McDermott AB, Fong Y, Benkeser D, Deng W, Zhou H, Houchens CR, Martins K, Jayashankar L, Castellino F, Flach B, Lin BC, O'Connell S, McDanal C, Eaton A, Sarzotti-Kelsoe M, Lu Y, Yu C, Borate B, van der Laan LWP, Hejazi NS, Huynh C, Miller J, El Sahly HM, Baden LR, Baron M, De La Cruz L, Gay C, Kalams S, Kelley CF, Andrasik MP, Kublin JG, Corey L, Neuzil KM, Carpp LN, Pajon R, Follmann D, Donis RO, Koup RA, Immune Assays Team\$, Moderna, Inc. Team\$, Coronavirus Vaccine Prevention Network (CoVPN)/Coronavirus Efficacy (COVE) Team\$, United States Government (USG)/CoVPN Biostatistics Team\$ (2022) Immune correlates analysis of the mRNA-1273 COVID-19 vaccine efficacy clinical trial. *Science*. <https://doi.org/10.1126/science.abm3425>
- Guo BQ, Li HB, Yang LQ (2023) Incidence of myopericarditis after mRNA COVID-19 vaccination: A meta-analysis with focus on adolescents aged 12–17 years. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2023.05.049>
- Hashiba K, Taguchi M, Sakamoto S, Otsu A, Maeda Y, Ebe H, Okazaki A, Harashima H, Sato Y (2024) Overcoming thermostability challenges in mRNA-lipid nanoparticle systems with piperidine-based ionizable lipids. *Communications Biology*. <https://doi.org/10.1038/s42003-024-06235-0>
- Hou X, Zaks T, Langer R, Dong Y (2021) Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials*. <https://doi.org/10.1038/s41578-021-00358-0>
- Hviid A, Nieminen TA, Pihlström N, Gunnes N, Dahl J, Karlstad Ø, Gulseth HL, Sundström A, Husby A, Hansen JV, Ljung R, Hovi P (2024) Booster vaccination with SARS-CoV-2 mRNA vaccines and myocarditis in adolescents and young adults: a Nordic cohort study. *European Heart Journal*. <https://doi.org/10.1093/eurheartj/ehae056>
- Hồ NT, Hughes SG, Ta VT, Phan LT, Đỗ Q, Nguyễn TV, Phạm ATV, Thị Ngọc Đăng M, Nguyễn LV, Trịnh QV, Phạm HN, Chữ MV, Nguyễn TT, Lương QC, Tường Lê VT, Nguyễn TV, Trần LTL, Thi Van Luu A, Nguyen AN, Nguyen NTH, Vu HS, Edelman JM, Parker S, Sullivan B, Sullivan S, Ruan Q, Clemente B, Luk B, Lindert K, Berdieva D, Murphy K, Sekulovich R, Greener B, Smolenov I, Chivukula P, Nguyễn VT, Nguyen XH (2024) Safety, immunogenicity and efficacy of the self-amplifying mRNA ARCT-154 COVID-19 vaccine: pooled phase 1, 2, 3a and 3b randomized, controlled trials. *Nature Communications*. <https://doi.org/10.1038/s41467-024-47905-1>
- Ioannou GN, Berry K, Rajeevan N, Li Y, Yan L, Huang Y, Lin HM, Bui D, Hynes DM, Rowneki M, Bohnert A, Boyko EJ, Iwashyna TJ, Maciejewski ML, Smith VA, Berkowitz TSZ, O'Hare AM, Vigilanti EM, Aslam M, Bajema KL (2025) Effectiveness of the 2023-to-2024 XBB.1.5 COVID-19 Vaccines Over Long-Term Follow-up. *Annals of Internal Medicine*. <https://doi.org/10.7326/annals-24-01015>
- Johnston TS, Li SH, Painter MM, Atkinson RK, Douek NR, Reeg DB, Douek DC, Wherry EJ, Hensley SE (2024) Immunological imprinting shapes the specificity of human antibody responses against SARS-CoV-2 variants. *Immunity*. <https://doi.org/10.1016/j.immuni.2024.02.017>
- Kandinov B, Soens M, Huang W, Llapur C, Ens D, Essink B, Fierro C, Vakili J, Pucci A, Guo J, Rudden S, Hall K, Hicks B, Schaefer K, Zhou H, Ma C, Zheng L, Avanesov A, Park Y, Du E, Miller J, Ananworanich J, Nachbagauer R (2025) An mRNA-based seasonal influenza vaccine in adults: Results of two phase 3 randomized clinical trials and correlate of protection analysis of hemagglutination inhibition titers. *Human Vaccines & Immunotherapeutics*. <https://doi.org/10.1080/21645515.2025.2484088>

47 · VERIFIED VIA CROSSREF

- Karikó K, Buckstein M, Ni H, Weissman D (2005) Suppression of RNA Recognition by Toll-like Receptors: The Impact of Nucleoside Modification and the Evolutionary Origin of RNA. *Immunity*. <https://doi.org/10.1016/j.immuni.2005.06.008>
- Karikó K, Muramatsu H, Ludwig J, Weissman D (2011) Generating the optimal mRNA for therapy: HPLC purification eliminates immune activation and improves translation of nucleoside-modified, protein-encoding mRNA. *Nucleic Acids Research*. <https://doi.org/10.1093/nar/gkr695>
- Karlstad Ø, Hovi P, Husby A, Härkänen T, Selmer RM, Pihlström N, Hansen JV, Nohynek H, Gunnes N, Sundström A, Wohlfahrt J, Nieminen TA, Grünwald M, Gulseth HL, Hviid A, Ljung R (2022) SARS-CoV-2 Vaccination and Myocarditis in a Nordic Cohort Study of 23 Million Residents. *JAMA Cardiology*. <https://doi.org/10.1001/jamacardio.2022.0583>
- Kim W, Zhou JQ, Horvath SC, Schmitz AJ, Sturtz AJ, Lei T, Liu Z, Kalaidina E, Thapa M, Alsoussi WB, Haile A, Klebert MK, Suessen T, Parra-Rodriguez L, Mudd PA, Whelan SPJ, Middleton WD, Teeffey SA, Pusic I, O'Halloran JA, Presti RM, Turner JS, Ellebedy AH (2022) Germinal centre-driven maturation of B cell response to mRNA vaccination. *Nature*. <https://doi.org/10.1038/s41586-022-04527-1>
- Klein NP, Lewis N, Goddard K, Fireman B, Zerbo O, Hanson KE, Donahue JG, Kharbada EO, Naleway A, Nelson JC, Xu S, Yih WK, Glanz JM, Williams JTB, Hambidge SJ, Lewin BJ, Shimabukuro TT, DeStefano F, Weintraub ES (2021) Surveillance for Adverse Events After COVID-19 mRNA Vaccination. *JAMA*. <https://doi.org/10.1001/jama.2021.15072>
- Kurhade C, Zou J, Xia H, Liu M, Chang HC, Ren P, Xie X, Shi P (2023) Low neutralization of SARS-CoV-2 Omicron BA.2.75.2, BQ.1.1 and XBB.1 by parental mRNA vaccine or a BA.5 bivalent booster. *Nature Medicine*. <https://doi.org/10.1038/s41591-022-02162-x>
- Lamoot A, Lammens J, De Lombaerde E, Zhong Z, Gontsarik M, Chen Y, De Beer TRM, De Geest BG (2023) Successful batch and continuous lyophilization of mRNA LNP formulations depend on cryoprotectants and ionizable lipids. *Biomaterials Science*. <https://doi.org/10.1039/d2bm02031a>
- Li C, Lee A, Grigoryan L, Arunachalam PS, Scott MKD, Trisal M, Wimmers F, Sanyal M, Weidenbacher PA, Feng Y, Adamska JZ, Valore E, Wang Y, Verma R, Reis N, Dunham D, O'Hara R, Park H, Luo W, Gitlin AD, Kim P, Khatrri P, Nadeau KC, Pulendran B (2022) Mechanisms of innate and adaptive immunity to the Pfizer-BioNTech BNT162b2 vaccine. *Nature Immunology*. <https://doi.org/10.1038/s41590-022-01163-9>
- Liu X, Li Z, Li X, Wu W, Jiang H, Zheng Y, Zhou J, Ye X, Lu J, Wang W, Yu L, Li Y, Qu L, Wang J, Li F, Chen L, Wu L, Feng L (2024) A single-dose circular RNA vaccine prevents Zika virus infection without enhancing dengue severity in mice. *Nature Communications*. <https://doi.org/10.1038/s41467-024-53242-0>
- Locci M, Pardi N (2026) Immunological mechanisms of mRNA vaccines for infectious diseases. *Nature*. <https://doi.org/10.1038/s41586-026-10599-0>
- Ma C, Du J, Lan L, Kapoor A, Marc GP, Jimenez G, Duncan CJA, Le Cam N, Lin N, Priddy F, Garg S, Stoszek SK, Shaw CA, Goswami J, Wilson E, Das R, Zhou H, Zheng L (2025) Immune correlates analysis of mRNA-1345 RSV vaccine efficacy clinical trial. *Nature Communications*. <https://doi.org/10.1038/s41467-025-61153-x>
- Ma H, Chen YY, Shih WL, Chen YC, Chen TJ, Fang CT (2026) Effectiveness of the monovalent XBB.1.5 COVID-19 vaccines: A systematic review and meta-analysis. *Journal of Microbiology, Immunology and Infection*. <https://doi.org/10.1016/j.jmii.2025.07.002>
- Meulewaeter S, Nuytten G, Cheng MHY, De Smedt SC, Cullis PR, De Beer T, Lentacker I, Verbeke R (2023) Continuous freeze-drying of messenger RNA lipid nanoparticles enables storage at higher temperatures. *Journal of Controlled Release*. <https://doi.org/10.1016/j.jconrel.2023.03.039>
- Mudd PA, Minervina AA, Pogorelyy MV, Turner JS, Kim W, Kalaidina E, Petersen J, Schmitz AJ, Lei T, Haile A, Kirk AM, Mettelman RC, Crawford JC, Nguyen THO, Rowntree LC, Rosati E, Richards KA, Sant AJ, Klebert MK, Suessen T, Middleton WD, Estep JH, Schultz-Cherry S, McGargill MA, Gaur A, Hoffman J, Mori M, Tang L, Tuomanen E, Webby R, Hayden RT, Hakim H, Hijano DR, Allison KJ, Allen EK, Bajracharya R, Awad W, Van de Velde LA, Clark BL, Wilson TL, Souquette A, Castellaw A, Dallas RH, Gowen A, Fabrizio TP, Lin CY, Brice DC, Cherry S, Roubidoux EK, Cortez V, Freiden P, Wohlgemuth N, Whitt K, Wolf J, Teeffey SA, O'Halloran JA, Presti RM, Kedzierska K, Rossjohn J, Thomas PG, Ellebedy AH (2022) SARS-CoV-2 mRNA vaccination elicits a robust and persistent T follicular helper cell response in humans. *Cell*. <https://doi.org/10.1016/j.cell.2021.12.026>
- Mulrone TE, Pöyry T, Yam-Puc JC, Rust M, Harvey RF, Kalmar L, Horner E, Booth L, Ferreira AP, Stoneley M, Sawarkar R, Mentzer AJ, Lilley KS, Smales CM, von der Haar T, Turtle L, Dunachie S, Klenerman P, Thaventhiran JED, Willis AE (2024) N1-methylpseudouridylation of mRNA causes +1 ribosomal frameshifting. *Nature*. <https://doi.org/10.1038/s41586-023-06800-3>
- Muramatsu H, Lam K, Bajusz C, Laczkó D, Karikó K, Schreiner P, Martin A, Lutwyche P, Heyes J, Pardi N (2022) Lyophilization provides long-term stability for a lipid nanoparticle-formulated, nucleoside-modified mRNA vaccine. *Molecular Therapy*. <https://doi.org/10.1016/j.ymthe.2022.02.001>
- Oberhardt V, Luxemburger H, Kemming J, Schulien I, Ciminski K, Giese S, Csernalabics B, Lang-Meli J, Janowska I, Staniek J, Wild K, Basho K, Marinescu MS, Fuchs J, Topfstedt F, Janda A, Sogukpinar O, Hilger H, Stete K, Emmerich F, Bengsch B, Waller CF, Rieg S, Sagar, Boettler T, Zoldan K, Kochs G, Schwemmle M, Rizzi M, Thimme R, Neumann-Haefelin C, Hofmann M (2021) Rapid and stable mobilization of CD8+ T cells by SARS-CoV-2 mRNA vaccine. *Nature*. <https://doi.org/10.1038/s41586-021-03841-4>
- Oda Y, Kumagai Y, Kanai M, Iwama Y, Okura I, Minamida T, Yagi Y, Kurosawa T, Greener B, Zhang Y, Walson JL (2024) Immunogenicity and safety of a booster dose of a self-amplifying RNA COVID-19 vaccine (ARCT-154) versus BNT162b2 mRNA COVID-19 vaccine: a double-blind, multicentre, randomised, controlled, phase 3, non-inferiority trial. *The Lancet Infectious Diseases*. <https://doi.org/10.1016/s1473-3099%2823%2900650-3>
- Pardi N, Krammer F (2024) mRNA vaccines for infectious diseases — advances, challenges and opportunities. *Nature Reviews Drug Discovery*. <https://doi.org/10.1038/s41573-024-01042-y>
- Patone M, Mei XW, Handunnetthi L, Dixon S, Zaccardi F, Shankar-Hari M, Watkinson P, Khunti K, Harnden A, Coupland CAC, Channon KM, Mills NL, Sheikh A, Hippisley-Cox J (2022) Risk of Myocarditis After Sequential Doses of COVID-19 Vaccine and SARS-CoV-2 Infection by Age and Sex. *Circulation*. <https://doi.org/10.1161/circulationaha.122.059970>
- Polack FP, Thomas SJ, Kitchin N, Absalon J, Gurtman A, Lockhart S, Perez JL, Pérez Marc G, Moreira ED, Zerbini C, Bailey R, Swanson KA, Roychoudhury S, Koury K, Li P, Kalina WV, Cooper D, Frenck RW, Hammitt LL, Türeci Ö, Nell H, Schaefer A, Ünal S, Tresnan DB, Mather S, Dormitzer PR, Şahin U, Jansen KU, Gruber WC (2020) Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2034577>
- Rosa SS, Prazeres DMF, Azevedo AM, Marques MPC (2021) mRNA vaccines manufacturing: Challenges and bottlenecks. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2021.03.038>
- Sahin U, Muik A, Derhovanessian E, Vogler I, Kranz LM, Vormehr M, Baum A, Pascal K, Quandt J, Maurus D, Brachtendorf S, Lörks V, Sikorski J, Hilker R, Becker D, Eller AK, Grütznier J, Boesler C, Rosenbaum C, Kühnle MC, Luxemburger U, Kemmer-Brück A, Langer D, Bexon M, Bolte S, Karikó K, Palanche T, Fischer B, Schultz A, Shi PY, Fontes-Garfias C, Perez JL, Swanson KA, Loschko J, Scully IL, Cutler M, Kalina W, Kyratsos CA, Cooper D, Dormitzer PR, Jansen KU, Türeci Ö (2020) COVID-19 vaccine BNT162b1 elicits human antibody and TH1 T cell responses. *Nature*. <https://doi.org/10.1038/s41586-020-2814-7>
- Song S, Madewell ZJ, Liu M, Miao Y, Xiang S, Huo Y, Sarkar S, Chowdhury A, Longini IM, Yang Y (2024) A systematic review and meta-analysis on the effectiveness of bivalent mRNA booster vaccines against Omicron variants. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2024.04.049>
- Tang J, Zeng C, Cox TM, Li C, Son YM, Cheon IS, Wu Y, Behl S, Taylor JJ, Chakraborty R, Johnson AJ, Shiavo DN, Utz JP, Reisenauer JS, Midthun DE, Mullon JJ, Edell ES, Alameh MG, Borish L, Teague WG, Kaplan MH, Weissman D, Kern R, Hu H, Vassallo R, Liu SL, Sun J (2022) Respiratory mucosal immunity against SARS-CoV-2 after mRNA vaccination. *Science Immunology*. <https://doi.org/10.1126/sciimmunol.add4853>
- Turner JS, O'Halloran JA, Kalaidina E, Kim W, Schmitz AJ, Zhou JQ, Lei T, Thapa M, Chen RE, Case JB, Amanat F, Raueo AM, Haile A, Xie X, Klebert MK, Suessen T, Middleton WD, Shi PY, Krammer F, Teeffey SA, Diamond MS, Presti RM, Ellebedy AH (2021) SARS-CoV-2 mRNA vaccines induce persistent human germinal centre responses. *Nature*. <https://doi.org/10.1038/s41586-021-03738-2>
- Verbeke R, Hogan MJ, Loré K, Pardi N (2022) Innate immune mechanisms of mRNA vaccines. *Immunity*. <https://doi.org/10.1016/j.immuni.2022.10.014>
- Wilson E, Goswami J, Baqui AH, Doreski PA, Perez-Marc G, Zaman K, Monroy J, Duncan CJA, Ujije M, Rāmet M, Pérez-Breva L, Falsey AR, Walsh EE, Dhar R, Wilson L, Du J, Ghaswalla P, Kapoor A, Lan L, Mehta S, Mithani R, Panozzo CA, Simorellis AK, Kuter BJ, Schödel F, Huang W, Reuter C, Slobod K, Stoszek SK, Shaw CA, Miller JM, Das R, Chen GL (2023) Efficacy and Safety of an mRNA-Based RSV PreF Vaccine in Older Adults. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2307079>